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Ministry for Primary Industries
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Attention: Kara McKelvey

INTERIM SET NET MEASURES TO MANAGE THE RISK OF MAUI'S DOLPHIN MORTALITY

The Environmental Defence Society (EDS) welcomes the opportunity to comment on the Ministry for Primary Industries' discussion paper on interim set net measures to manage the risk of Maui's dolphin mortality (the discussion paper).

Given the critical status of the Maui's dolphin population, EDS considers that action should be taken as a matter of urgency to minimise the risk to Maui's dolphins from fishing activity.

BACKGROUND TO EDS

EDS is a public interest environmental organisation, formed in 1971, with a membership consisting largely of resource management professionals. The focus of EDS's work is on achieving good environmental outcomes through improving the quality of New Zealand's legal and policy frameworks and statutory decision-making processes. EDS both litigates in favour of the environment and operates as an environmental think tank.

EDS has had a long interest in the management of New Zealand's coast and oceans and in marine conservation more generally.

- In 2005 EDS published a report on oceans governance titled *Looking out to sea: New Zealand as a model for ocean governance*.
- In 2007 EDS published a report investigating integrated coastal management titled *Beyond the tide: Integrating the management of New Zealand's coasts*. The report was reprinted in 2009 with support from the Hauraki Gulf Forum.
- In 2009, EDS released a policy paper on the establishment of an Environmental Protection Authority (EPA) titled *Improving environmental governance: the role of an Environmental Protection Authority*. This paper canvassed gaps in current marine management (amongst other things) and proposed a role for the EPA which would include oversight of coastal and marine management. It also proposed the establishment of a Coastal Commission.
- In 2009 EDS also launched a comprehensive book on coastal management titled *Castles in the Sand: What's happening to the New Zealand coast?*
- In April 2010 EDS launched a comprehensive guide to how the New Zealand's marine area is currently managed titled *Managing the Marine Environment*.
- In April 2011 EDS published a policy paper titled *Governing our Oceans: Environmental Reform for the Exclusive Economic Zone* which analysed the deficiencies in existing law and called for

urgent environmental effects legislation for the EEZ, given the growing interest in new commercial activities in the marine area.

- EDS is involved in ongoing work on coastal and marine conservation and management issues. In particular, EDS is currently undertaking research into the adequacy of the current legal and policy framework for managing human impacts on marine mammals.

SUMMARY OF EDS'S SUBMISSIONS

A summary of EDS's submissions is outlined below:

1. EDS supports the proposal to adopt interim measures to extend the recreational and commercial set net ban into the area from Pariokariwa Point to Hawera out to four nautical miles but considers that in light of the extremely critical state of the Maui's dolphin population, a strongly precautionary approach is required and thus the interim measures should go much further.
2. EDS considers that interim measures should prohibit the use of set nets and trawl nets in all West Coast North Island waters with depths of less than 100 metres including within harbours.
3. EDS urges the Ministry for Primary Industries to develop permanent measures to protect the Maui's dolphin from fishing related mortality and to ensure that the Threat Management Plan is reviewed as a matter of urgency.

LEGISLATIVE FRAMEWORK

Fisheries Act 1996

The proposed set net ban is to be implemented under the Fisheries Act 1996. Under that Act the Ministry for Primary Industries has a duty to ensure that dependent species such as Maui's dolphins are sustained above a level which ensures their long term viability.

The purpose of the Fisheries Act includes *[a]voiding, remedying, or mitigating any adverse effects of fishing on the aquatic environment*. In making decisions under the Act, the Minister must take account of environmental principles including the following:

- (a) Associated or dependent species should be maintained above a level that ensures their long-term viability;*
- (b) Biological diversity of the aquatic environment should be maintained;*

"Associated and dependent species" are defined in the Act as *"any non-harvested species taken or otherwise affected by the taking of any harvested species"*. It is clear that the Maui's dolphin is an associated and dependent species given the fact that it is known to be caught in nets.

International obligations

The Ministry also has obligations to safeguard the Maui's dolphin population under international law.

Section 5(a) of the Fisheries Act 1996 (the Act) requires that the Act should be interpreted in a manner consistent with New Zealand's international obligations relating to fishing. Of relevance is the following provision under the United Nations Convention on the Law of the Sea of 10 December 1982:

Article 61 requires coastal states to ensure that, through proper conservation and management measures, that the maintenance of living resources in the EEZ is *"not endangered by over exploitation."* In taking such measures *"the coastal State shall take into consideration the effects on species associated with or dependent upon harvested species with a view to maintaining or restoring*

populations of such associated or dependent species above levels at which their reproduction may be seriously threatened."

Implementation under section 11

It is proposed that an interim extension to the set net ban will be implemented pursuant to section 11 of the Fisheries Act 1996. It is noted in the discussion paper that sections 11, 15 or 16 could be used to implement interim measures. Presumably the Ministry has received legal advice as to which of the three sections of the Fisheries Act should be applied.

It is noted that none of these provisions is a particularly appropriate in the present case. This highlights the need for fundamental reform of the law in relation to the protection of marine mammals so that decisions such as these may be taken under legislation which is designed for the purpose.

NEED FOR INTERIM MEASURES

Population status

It is indisputable that the Maui's dolphin population is in a highly critical state. It is listed as 'Nationally Critical' by the Department of Conservation. The *IUCN Red List of Threatened Species* classifies it as "critically endangered." The only IUCN categories denoting more serious circumstances are "extinct in the wild" and "extinct".

The current classifications are based on evidence of a catastrophic decline in the population over three generations since 1970 and the small number of individuals remaining. It is estimated that there would have been around 577 Maui's dolphins in 1970, and around 135 in 1985.¹ Data collected between 2001 and 2007 indicated that there were around 80 dolphins remaining.

The Department of Conservation's 2012 study *Estimating the abundance and effective population size of Maui's dolphins using microsatellite genotypes in 2010-11, with retrospective matching to 2001-7* (Hamner et al)² indicates a current population size of between 48 and 69 adult individuals.

Hamner et al note that although they did not find conclusive evidence of a decline in the population between 2001 and 2012, the *'analysis does suggest that a small decline is likely. It is important to note that the power to detect a decline decreases as population size decreases...and that [the] results do not offer conclusive evidence that the population is not declining.'*³ In addition, the Ministry for Primary Industries' preliminary assessment estimates that there is a 75% likelihood that the population is declining.

Despite the dire figures, there are still some reasons to be hopeful that, given good management, the population can recover. Hamner et al found that there is still a relatively large number of females within the population, and a relatively high level of genetic diversity.

In addition, the identification of two female Hector's dolphins amongst the Maui's dolphins provides the possibility that Hector's and Maui's may successfully interbreed. Whilst there are a number of reasons that this may not be the solution, the severity of the situation means that the importance of such opportunities must be recognised.

For this reason, the fact that the dolphin that was caught in a commercial set net off Taranaki was not positively identified as either a Hector's dolphin or a Maui's dolphin is not critical in relation to whether a set net ban in the area would benefit the Maui's dolphin: either the dolphin was a Maui's, in

¹ Burkhart S M and E Slooten, 2003, 557

² Hamner, RM, Oremus, M Stanley, M Brown, P Constantine, R and Baker, C, 2012, *Estimating the abundance and effective population size of Maui's dolphins using microsatellite genotypes: 2010-11 with retrospective matching to 2001-7* Department of Conservation

³ Ibid p21

which case its loss represents a significant blow for the population, or it was a Hector's which could have been in a position to interbreed with Maui's.

Distribution

In the discussion paper it is stated that "*the Taranaki area is not considered to be part of the Maui's dolphin's core range. Dolphins are considered to be rare and infrequent visitors to the area.*"⁴ In fact the Taranaki-Whanganui region was historically a core part of the Maui's dolphin population's distribution: it is only as the population has declined that sightings of dolphins in this region have become more infrequent and the population has become centred on the area further north.

Although it was previously thought that Maui's dolphins have a restricted home range, Hamner et al show that Maui's dolphins travel much further than previously thought – the maximum recorded distance was 78 kilometres over 19 days. This latest discovery highlights the fact that there are gaps in our understanding of Maui's dolphin behaviour, which mean that we should apply a strongly precautionary approach in respect of measures to protect them.

There is strong evidence to suggest that the dolphins travel into the area in which the set net ban is proposed. Although the study by Hanmer et al did not record any sightings in the Taranaki region, there are a number of documented sightings in the region, including footage filmed on a cell phone by a fisherman in 2009. The death of a dolphin in a set net off Taranaki in January 2012 confirms that they are present there and are at risk of entanglement in set nets.

The presence of two Hector's dolphins off the west coast of the North Island indicates that they are able to travel from the South Island into the habitat of the Maui's dolphins. Thus as noted by Hanmer et al, the establishment of corridors within and between core distributions of Maui's and Hector's dolphins may be important for the survival of the Maui's population.⁵

Fishing related mortalities

The Ministry for Primary Industries' preliminary assessment of potential biological removal indicates that the population can sustain one human-induced mortality every ten to twenty three years without impacting on the ability of the population to rebuild to its optimum population size.

However, since 2002 twelve Maui's dolphin mortalities have been recorded in the Department of Conservation incident database. Three of these were confirmed deaths in set nets, including the mortality which occurred off Taranaki on 2 January 2012. A further three mortalities were probably or possibly caused by set nets. Drownings in set nets are thus the largest known cause of Maui's dolphin mortality.

The indiscriminate nature of set netting means that in many countries, including the United States, United Kingdom and Australia, it is prohibited or highly regulated. As illustrated by the January mortality in Taranaki, current measures to manage the risk of Maui's dolphin bycatch in set nets in New Zealand are inadequate. Whilst set netting is permitted in Maui's dolphin habitat a key threat to the survival of this severely threatened species will remain.

Although there have been no recorded mortalities in trawl nets on the West Coast of the North Island, it is highly likely that trawl fisheries are also a threat to the survival of the Maui's dolphin. Hector's dolphins are known to die as a result of entanglement in trawl nets. The absence of observers or video surveillance on fishing vessels means that it is possible that mortalities have occurred but not been reported. There is little incentive for fishers to report such incidents because by doing so they may jeopardise their ability to continue fishing.

⁴ Ministry of Agriculture and Forestry *Interim Set Net Measures to Manage the Risk of Maui's dolphin mortality* MAF Discussion Paper 2012/03 page 5

⁵ Hanmer et al page 22

CONCLUSION

Whilst entanglement in nets is unlikely to be the only threat to Maui's dolphins, it is the most significant known threat, and as such measures should be taken as a matter of urgency to ensure that no further Maui's dolphin mortalities occur as a result of fishing activity.

EDS supports the proposal to extend the set net ban to cover the area from Pariokariwa Point to Hawera. We recommend that as a minimum the boundaries should be aligned with those of the northern area, so that the ban area extends to seven nautical miles in all areas.

We urge the Minister to consider further extending bans on both set nets and trawl nets to all West Coast North Island waters out to the 100 metre depth contour and including within harbours as an interim measure.

EDS does not favour extreme responses to conservation challenges as a matter of course. It is recognised that interim fisheries closures will have an impact on the fishing industry on the west coast. However in this case we consider that the risk to the Maui's dolphin is so severe that far reaching action is required.

The recent discovery that dolphins travel further than previously thought highlights the fact that we do not have a full understanding of Maui's dolphin behaviour and distribution and therefore cannot be confident that limited measures focussed on the 'core' areas of Maui's dolphin habitat will be effective. The significance of losing even one more Maui's dolphin means that we need to ensure that protection measures are comprehensive, covering the entirety of the dolphins' range.

If the Minister is not able to take such a measure, we urge the Minister to consider implementing a requirement that all relevant fishing vessels carry independent observers, or at the very least, use video surveillance equipment, as a matter of urgency so that data can be gathered to ensure that permanent measures are targeted effectively.

Having put interim measures in place, it is important that the Ministry for Primary Industries and the Department of Conservation proceed with permanent measures to safeguard the future of the Maui's dolphin as a matter of urgency, including reviewing the Threat Management Plan.

EDS would welcome the opportunity to discuss our submission further with you.

Yours sincerely



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